

Final project – Spatial analysis

CitiBike trips in NYC and busy routes

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Problem statement:

In this project, I analyzed bike trips in NYC using CitiBike services to identify routes that are congested or heavily used, which may negatively impact user experience and contribute to road and sidewalk congestion in an already busy city.

For this analysis, I used data from CitiBike and NYC OpenData databases. Each month contains approximately 250,000 trips. Due to computational limitations and the need for a visually clear map, I randomly selected 250 rides from January 2019. This month was chosen arbitrarily as a pre-pandemic reference point.

It's important to say that I was able to show on a map the busy routes and Further analysis with more data is required to observe changes over time (month/day) and evaluate the impact of special events. Additionally, it is important to study how CitiBike usage patterns have changed post-COVID-19.

Assumptions:

- The users use the bike to go from point A to point B as a transportation option and not for exercise or deliveries and such.
- The rides are generally on bike paths and only for a few locations or rides isn't using the bike paths but the sidewalks or the roads.

Data:

The data I used is from CitiBike website and from NYCOpenData website as follows(the links are cited in the end of the document):

CitiBike :

- Dataset of the bike trips in NYC – the data specifically that is collected is from Manhattan, Queens and Brooklyn in a CSV file where each row is a bike trip with start station and stop station, when the stations are stationary so you can't just leave the bike anywhere.

NYCOpenData:

- Shapefiles of Borough Boundaries, Bike paths across NYC – the assumption is that the trips are.

A snippet of the table of bike trips

start station id	start station latitude	start station longitude	end station id	end station latitude	end station longitude
3160	40.77896784	-73.97374737	3283	40.7882213	-73.97041561
519	40.751873	-73.977706	518	40.74780373	-73.9734419
3171	40.78524672	-73.97667321	3154	40.77314236	-73.95856158
504	40.73221853	-73.98165557	3709	40.73804614	-73.99642959
229	40.72743423	-73.99379025	503	40.73827428	-73.98751968
3630	40.8038654	-73.9559308	3529	40.8107922	-73.9430681
3675	40.7845968	-73.94968539	3288	40.778301	-73.9488134
285	40.73454567	-73.99074142	3358	40.6711978	-73.97484126
3394	40.6769993	-74.00647134	3398	40.6746957	-73.99785768

The key columns are the longitude and latitude of the start and stop stations which are used in the route analysis to get the routes of each trip under the assumptions.

In the process of trying to do the analysis, I tried to use the time stamps and the type of users in each trip, but like mentioned above the time stamps are encoded and for the type of users I couldn't find any special details about each one of them or some kind of special behaviors\using patterns. Using more data may lead to pattern recognition.

Methodology:

1. Loading and preparing the Data:

- 1.1. Load the Citi Bike dataset into ArcGIS Pro.
- 1.2. Load the bike_routes and borough boundaries shapefiles
- 1.3. Convert Start and End Locations to Points using the XY Table to Point tool, ensuring all points are in the same coordinate system-WGS84(DD).

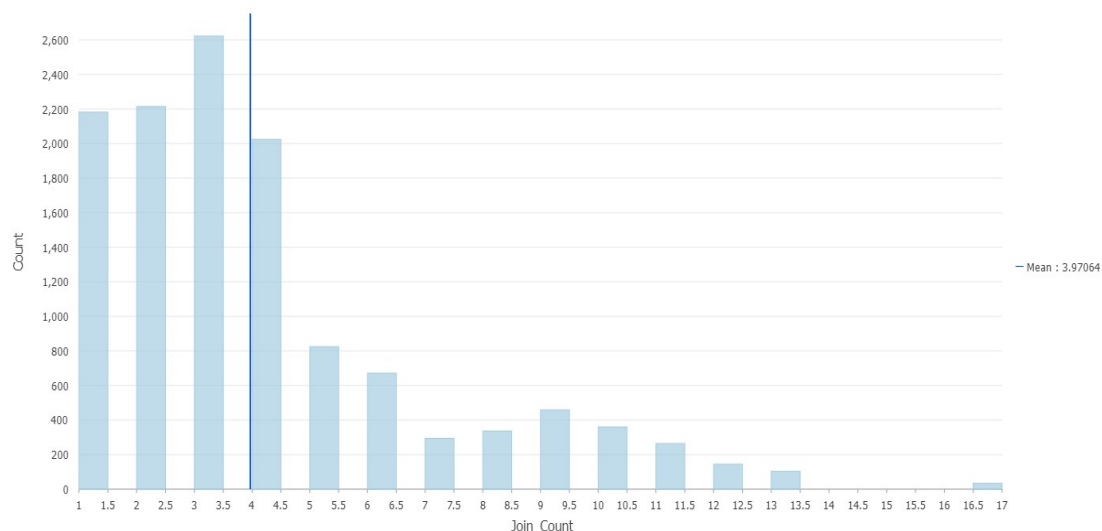
2. Create a Network Dataset for Bike Routes:

- 2.1. Convert the bike_routes layer into a Feature Class within a File Geodatabase (GDB) using feature class to Geodatabase.
- 2.2. Create a Network Dataset from the bike routes layer:
 - Create in the Network Dataset a Travel mode with the attributes that you need (I created without any specific restrictions and U-turn can be made everywhere because it is bike rides)

3. Perform Route Analysis:

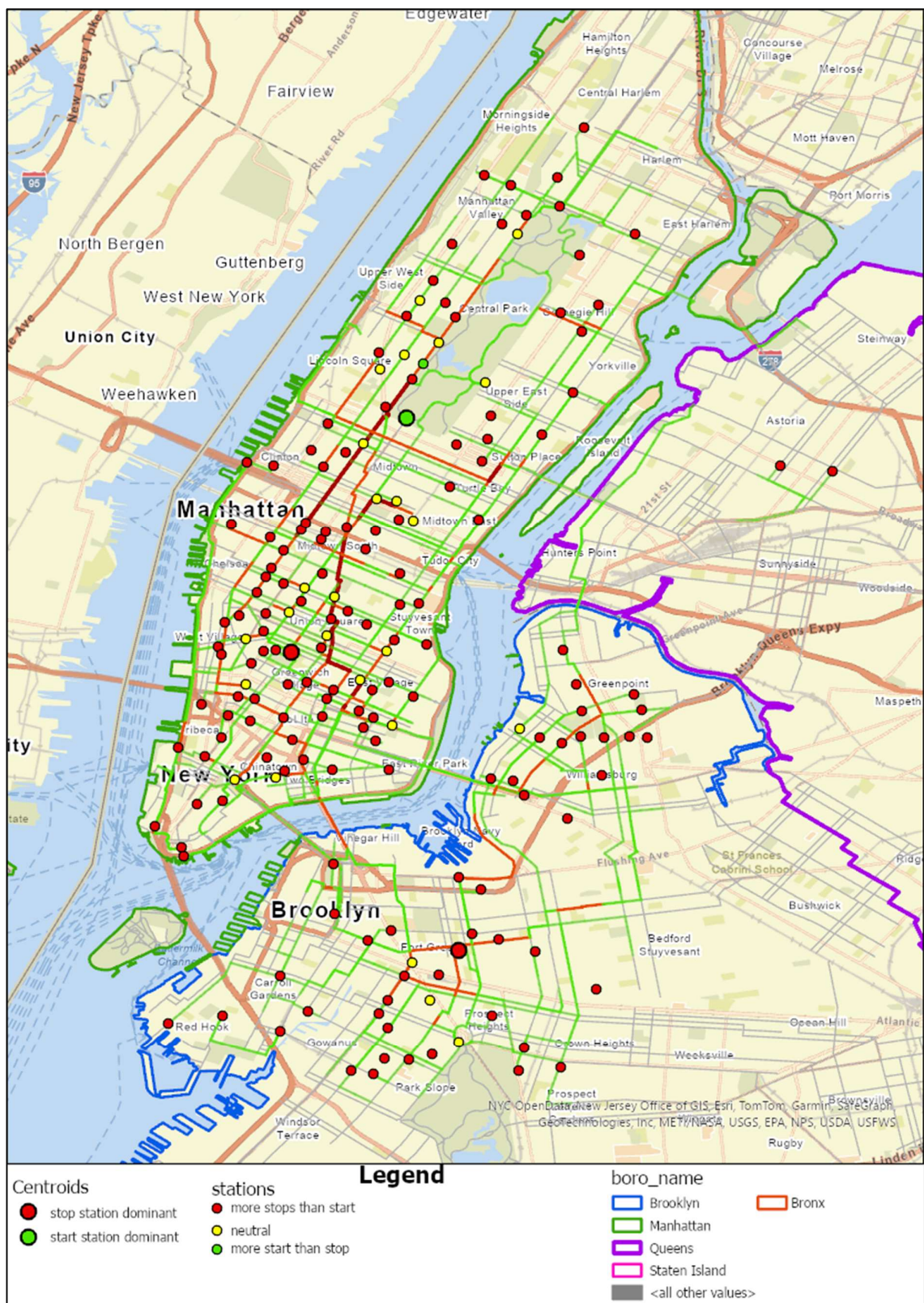
- Create a **Route Analysis Layer** using Network Analyst and select Bike_Network_ND as the dataset.
 - Load Start and End Points as Stops to establish the trip routes – it is important to that the first points are the start and after that the stop points because there is sequence of the route and 1 is the first station and 2 is the one after – the stop station.
 - Solve the network analysis to generate the shortest bike routes.
4. Self-join and using summary statistics to get for each station if it used more as a start or a stop point. Use symbology for red for more stops than start(0 and below), green for more
 5. Perform K-means clustering to identify key bike station clusters on the map. Then, calculate the mean center for each cluster. A value of K=3 was selected as the study area is roughly divided into three major regions.
 6. Use symbology to change the color of the centroids to red for more stops than starts and green otherwise.
 7. Intersect the junctions with the bike_paths of NYC to create segments of the roads – by using split line at point.
 8. Use spatial join on the split paths with the routes of the trips, using '**Share a line segment with**'.
 9. Open the table of the spatial join and with the symbology create 3 levels of load on the paths with the 'Join_Count' column – which counts how many routes cross this segment(low: 1-3 trips,medium: 4-7,high: 8-17).
 10. Also create a histogram to get how many routes have 1,2,3 and so on trips.

Histogram of the number of trips in each segment:



The Maps:





Interpretation of the Analysis: What Did I Find?

The analysis reveals that some bike routes in NYC experience significantly higher usage than others, leading to challenges for both cyclists and pedestrians. Overcrowded bike paths may force riders onto sidewalks or roads, increasing the risk of accidents.

The findings also reveal an imbalance in bike distribution. Some areas have more trips ending than starting, leading to potential bike shortages. Conversely, other locations experience an oversupply of bikes, making availability inconsistent. It is uncertain whether this imbalance is a long-term trend or specific to January. Additional data analysis over multiple months is necessary to determine if this pattern persists. Since the dataset was randomly sampled, it provides a general overview of bike usage and distribution, but validation with a larger dataset is necessary for more accurate conclusions.

The busiest CitiBike routes, shown in dark red on the map, include Broadway, 7th Avenue, 6th Avenue, 1st Avenue, 2nd Avenue, and the Hudson River Greenway. These routes have high bike traffic because they connect key transit hubs (such as Penn Station and Grand Central), business districts (like Midtown and the Financial District), and popular recreational areas (such as Central Park and the Hudson River Greenway). Additionally, the presence of protected bike lanes on streets like 1st and 2nd Avenues makes them preferred routes for cyclists, attracting commuters, tourists, and recreational riders.

Suggestions and Further Research:

City planners and decision-makers can use this data to enhance bike infrastructure by expanding bike lanes, increasing the number of docking stations, and adjusting traffic regulations. However, since this study is based on data from only one-month, further research is necessary to analyze how bike usage changes across different times of the day, days of the week, and seasons.

To address bike congestion and distribution imbalances, planners should consider developing dedicated bike "highways" in high-traffic areas. Introducing incentives, such as discounts or coupons (for example for stores near the stations), could encourage users to begin or end their trips at designated stations. This strategy may help distribute bikes more evenly across locations. These improvements would reduce overcrowding on the busiest routes, improve efficiency, and ensure a safer cycling experience for all users. Future research should also examine external factors that may influence bike usage, such as weather conditions, major events, and policy changes.

Links:

Bike_paths:

https://data.cityofnewyork.us/dataset/New-York-City-Bike-Routes/mzxc-pwib/about_data

NYC department of transportation:

<https://www.nyc.gov/html/dot/html/bicyclists/bicyclists.shtml>

boroughs borders:

<https://www.arcgis.com/home/item.html?id=1f75b59e45444d7bae27454ec7730395>

(The original link of boroughs borders is down from NYC site

<https://data.cityofnewyork.us/City-Government/Borough-Boundaries/tqmj-j8zm>)

CitiBike trips:

<https://citibikenyc.com/system-data>